

Module 10: infrastructure Security an WAN Technologies-

13- A Cisco Catalyst switch connects to what should be individual user PCs. Each port has the same port security configuration, configured as follows:
interface range gigabit Ethernet 0/1 – 24

switchport mode access

switchport port-security

switchport port-security mac-address sticky

Which of the following answers describe the result of the port security configuration created with these commands? (Choose two)

A. Prevents unknown devices with unknown MAC addresses from sending data through the switch ports.

B. If a user connects a switch to the cable, prevents multiple devices from sending data through the port.

C. Will allow any one device to connect to each port, and will save that device's MAC address into the startup-config

D. Will allow any one device to connect to each port, but will not save that device's MAC address into the startup-config

A-[If another unknown device connects the switch will block traffic from the device]

C- [sticky will store the learned Mac address in running config (and in startup config if save) so practically only one PC can connect per port then other will deny]

14- What is the Administrative Distance of internal EIGRP routes?

A. 170

B. 90

C. 20

D. 1

E. 110

F. 120

[Administrative distance is the value that shows how trustworthy a routing protocol is. A lower value means more trusted. Internal EIGRP has an administrative Distance of 90, so it can be considered more reliable than protocols with higher values like OSPF.]

15- When a subnet mask is presented in binary, what do the binary 1s represent?

A. The network portion of an associated address

B. The host portion of the subnet mask

C. The number of wildcard bits in the subnet mask

D. The number of wildcard bits in the address

E. The network portion of the subnet mask

F. The host portion of an associated address

[In a subnet mask 1s = network part, 0s = host part. Thus, 1s show which parts are a part of the network in an IP address.]

16- Which switch would STP choose to become the root bridge in the selection process?

A. 32768: 11-22-33-44-55-66

B. 32768: 22-33-44-55-66-77

C. 32769: 11-22-33-44-55-65

D. 32769: 22-33-44-55-66-78

STP selects the root bridge based on the lowest bridge ID. Bridge ID includes priority and MAC address. The switch with the lowest priority value is selected. If the priority is the same, the switch with the lowest MAC address is selected. Option A has the lowest overall bridge ID.

17- Which of the following devices is used by the service provider to provide WAN services?

- A. Router
- B. Core router
- C. WAN switch
- D. CSU/DSU IOS router is

[A WAN switch is used inside the service provider network to move traffic between customers across wide area networks it is the part of the provider infrastructure]

18- Your Cisco acting as a DHCP server. Which command will display the addresses that have been handed out to clients on the LAN?

- A. show ip dhcp assignments
- B. show ip dhcp address
- C. show ip dhcp conflicts
- D. show ip dhcp bindings
- E. show ip dhcp pool

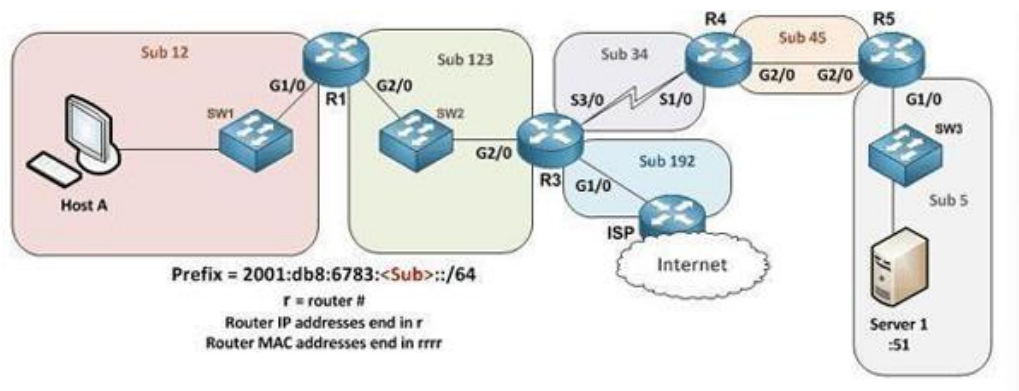
[The command shows ip dhcp building display the list of the IP addresses that have been assigned to client by dhcp server along with their Mac addresses and lease information]

19- Which of the following commands would you use to enable EIGRP only on those interfaces with an IP address from 10.1.1.0 through 10.1.1.63?

- A. network 10.1.1.0 0.0.0.63
- B. network 10.1.1.0/63
- C. router eigrp 10.1.1.0 0.0.0.63
- D. network 10.0.0.0 0.0.0.255

[EIGRP uses wildcraft masks in the network command. The wildcraft mask 0.0.0.63 matches with the specific IP range from 10.1.1.0 to 10.1.1.63. Therefore, this command enables EIGRP only on interface within that range.]

20- R3 has a static route configured that points toward the service provider. What command could you use to have R3 advertise an OSPFv3 default route to the internal network, regardless of whether R3 had its default static route?



- A. The decision to advertise a default route depends on the static route always being present on R3.
- B. The default behaviour is to redistribute any default IPv6 routes into OSPFv3, so no action is required.
- C. Each of the other routers needs a static default route that leads to R3.
- command
- D. Use the default-information originate always in interface mode for G1/0 on R3.
- E. Have R3 use the command default-information originate always in OSPFv3 router configuration mode.

[The command default information originated always in OSPDF V3 router configuration more force the router to advertise a default router to other routers, even if default static router does not currently exist.]

21- You are configuring dynamic NAT on your Cisco IOS router. Which command is used to verify the interfaces that are being used as the outside interface and the inside interface?

- A. show interfaces
- B. show ip route
- C. show ip nat translations
- D. show ip interface brief
- E. show ip interface
- F. show ip nat statistics

[The command shows IP interface display detailed information about each interface, including whether it is configured as nat inside or nat outside]

22- When using the “show EtherChannel summary “command, what does the “u” flag signify?

- A. Waiting to be aggregated
- B. Suspended
- C. In use
- D. Unsuitable for bundling

[In the EtherChannel summary output the flag “U” means the ether channel is up and functioning correctly. it includes that the bundles links are operational and actively forwarding traffic]

23- Which command could you enter to encrypt passwords?

- A. enable secret
- B. username {username} secret {password}
- C. service password-encryption
- D. All the above
- E. None of the above setting up a

[enable Secret stores the enable password in encrypted form. username secret store user password in encryption form. service password-encryption encrypt plain-text password in the configuration all this command provides password encryption, so the correct answer is all about.]

24- You are Cisco IOS router as a DHCP server. Which command is used to identify the IPv4 addresses that will be in the DHCP pool?

- A. network
- B. dns-server
- C. default-router
- D. ip dhcp excluded address
- E. lease

- F. ip dhcp pool
- G. domain-name

[The network command inside dhcp pool configuration define the network and subnetting mask for IP addresses that will be assigned to client]

25- Which of the following statements are true regarding the processing of ACLs that have been applied to router interfaces? (Choose two)

- A. Inbound ACLs will be processed before the routing table lookup occurs
- B. Inbound ACLs will be processed after the routing table lookup has occurred
- C. Outbound ACLs will be processed after the routing table lookup has occurred
- D. Outbound ACLs will be processed before the routing table lookup occurs

[Inbound ACLs are checked before the router perform routing table lookup. Outbound ACLs are checked after the router make the routing decision and select the outgoing interface]

26- imagine you configured OSPFv2 in a small lab network. Which of the following answers list a condition that could keep the routers in your lab from learning all the routes to all the IPv4 routes in your small lab network? (Choose two)

- A. An ACL could be blocking router advertisements.
- B. Two neighbouring routers that connect to the same link have been configured with the same OSPF area and with the same IPv4 subnet mask.
- C. Any physical layer problem that would prevent two neighbouring routers from being able to ping each other's IPv4 addresses in the subnet that exists between the two routers.
- D. Two neighboring routers that connect to the same link have been configured with the same OSPF process ID on the router ospf command.

[An ACL blocks OSPF traffic, router cannot form neighbour relationship, so routes will not be able exchanged.

if there is a physical layer issue, router cannot communicate, which prevent adjacency formation and route learning.]

27- Which statements describe neighbor discovery functionality in IPv6?
(Choose two)

- A. Determines the link layer address of a neighbor
- B. Finds neighbor switches on the link
- C. Is achieved by using Dynamic Host Configuration Protocol for IPv6, or DHCPv6 with IPv6 multicast
- D. Queries for duplicate addresses

[Neighbour discovery in IPV 6 determines the link-layer address of the neighbouring device and perform duplicate address detection to ensure that an IPv6 address is not already in use]

28- Which IPv6 prefix will the typical enterprise network receive from the service provider?

- A. /52
- B. /56
- C. /64
- D. /32
- E. /48
- F. /60

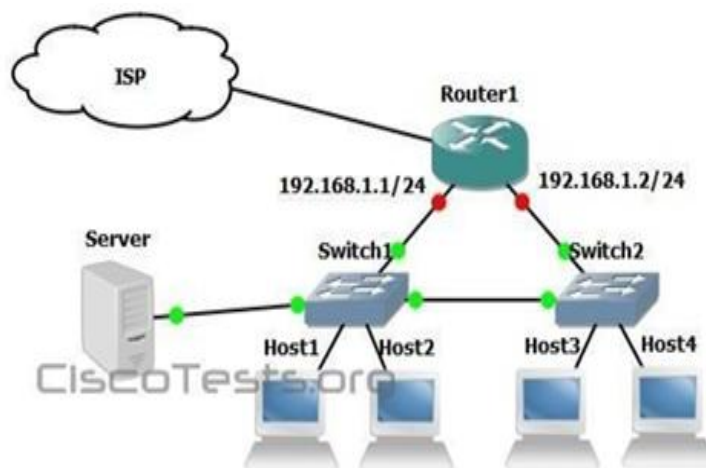
[A/48 prefix is a typically assigned to enterprise network by service providers. it provides many subnetting of for internal use]

29- How should be configured a switch so that it could be accessed remotely?

- A. Assign a password and privilege level
- B. Apply the access control list, or ACL, to the virtual type of terminal, or vty, lines
- C. Configure a gateway for the switch
- D. Generate a certificate

[For Remote access, a switch must have a configured default gateway so it can communicate which device outside is local network.]

30- Refer to the exhibit. A network technician is asked to design a small network with redundancy. The exhibit represents this design, with all hosts configured in the same VLAN. What conclusions can be made about this design?



- A. This design will function as intended.
- B. Spanning-tree will need to be used.
- C. The router will not accept the addressing scheme. connection
- D. The between switches should be a trunk.
- E. The router interfaces must be encapsulated with the 802.1Q protocol.

[Snapping Tree protocol is required to prevent switching loops when redundant links exists. The link between switches should be configured as a trunk to properly carry vlan traffic between switch]